

Multi-UAV Telecommunication Inspection against Intelligent Jamming Attacks: A Reinforcement Learning Framework

1st Bingzhi Zeng

School of Cyber Science and Engineering
Southeast University
Nanjing, China
bingz@seu.edu.cn

2nd Xin Gong*

School of Cyber Science and Engineering
Southeast University
Nanjing, China
xingong@seu.edu.cn

Abstract—This paper designs a scenario for unmanned aerial vehicles (UAVs) to assist in telecommunication tower inspection. There are opportunities for further exploration in several areas: 1) Comprehensive practical constraints should be considered; 2) Strong or intelligent attacks should be considered; 3) Single-UAV scenarios should not be the only scenarios considered. This paper focuses on a task where a swarm of UAVs fly within a certain space to inspect the signal strength of telecommunication towers, while also introducing intelligent interference attacks. Specifically, the UAV swarm needs to use a path planning algorithm with excellent robustness to counter this intelligent jamming attack, complete inspection from multiple telecommunication towers, and reach the destination within a specified time limit, while satisfying various practical constraints, including kinematic constraints, no-fly zone constraints, and task deadline constraints. This paper adopts the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm from Multi-Agent Reinforcement Learning (MARL) to solve this problem. This work gradually illustrates the effectiveness and robustness of defense measures in scenarios with no jamming attacks, fixed jamming attacks, and intelligent jamming attacks.

Index Terms—*jamming attacks, multi-agent deep reinforcement learning, path planning, UAV*

I. INTRODUCTION

Unmanned aerial vehicles (UAVs), also known as UAVs, are aircraft controlled remotely or autonomously without a human pilot onboard. UAVs possess unique characteristics, including flexibility and maneuverability, making them a popular research topic with potential for widespread applications in the coming decade [1]. Specifically, UAVs can be effectively utilized in intelligent transportation [2], intelligent detection [3], and auxiliary communication [4] scenarios. As noted in [5], UAVs offer advantages such as higher efficiency, lower

cost, and greater safety in the inspection of telecommunication towers. However, the broadcast nature of wireless transmissions makes this auxiliary communication scenario susceptible to interference attacks [6], particularly when the jamming signal originates from a mobile entity [7]. Therefore, developing a path planning algorithm for UAVs with resilience against jamming attacks is of great significance. This problem presents several challenges: 1) the uncertainty of the mobile jammer's information, 2) the resulting time-varying and non-stationary environment, and 3) the need to consider multiple practical constraints. In this paper, this article employs a MARL algorithm to obtain an effective elastic path planning algorithm for this scenario and propose an effective anti-jamming strategy.

Similar scenarios have been studied. In [8], a decisive flight path planning for UAS to visually inspect a power transmission line and towers is explained, using particle swarm optimization and simulated annealing to find the best path for UAS, but it didn't consider any attacks. In [9], a jammer location algorithm based on received signal strength to help UAV plan its path was proposed. In [10], an iterative algorithm was proposed to solve the 3D anti-interference trajectory design problem of UAV by using block coordinate descent and successive convex approximation technique. In [11], the author considered multiple ground jammers in the multi-UAV path planning problem, and proposed two algorithms based on block coordinate descent to obtain sub-optimal solutions with the help of relaxation variables, successive convex approximation technique and S-procedure. The above studies considered interference attacks from the ground, and most of them used traditional optimization methods to solve the path planning problem for UAVs. Deep reinforcement learning has attracted wide attention in solving highly complex optimization problems that cannot be handled by traditional mathematical methods. For example, resource allocation in V2V communication [12] and stochastic shortest path issues [13]. In [14], an automatic flight control algorithm based on reinforcement learning was proposed. However, jamming attack in this scenario was still unintelligent. In [15], authors

*Corresponding Author

This work was supported by the National Natural Science Foundation of China under grants 62003374 and 62403128, the Fundamental Research Funds for the Central Universities under Grant 08002150138, the Hunan Provincial Natural Science Foundation of China under grant 2020JJ5765, the Jiangsu Provincial Natural Science Foundation of China under Grants No. BK20210223 and BK20241284, the Nanjing Science and Technology Innovation Project for Overseas Scholars under Grant 4209012304, and Start-up Research Fund of Southeast University under Grant RF1028623260.

proposed an elastic path planning algorithm based on D3QN, and introduced intelligent jamming attacks, that is, jammers with learning ability. However, authors only considered the situation of a single UAV and a single intelligent jammer, and adopts discrete action space, which lacked the ability to adapt to the cooperation of multiple UAVs in the actual scene. In some cases, a single UAV cannot complete complex tasks, so a UAV group consisting of multiple UAVs is often used in cellular communications [16]. It is worth noting that the algorithm in the multi-agent scenario is not a simple extension of the independent training of each agent. In this case, from any agent's point of view, the environment will become unstable as agents independently update their policies. The non-stationary Markov process cannot guarantee the convergence of the algorithm.

This paper proposes the use of the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm [17] to address cooperative anti-jamming and path planning problems in multi-UAV collaboration scenarios, considering various practical constraints. The algorithm demonstrates good performance and scalability in multi-agent environments [18]. The main contributions of this paper are as follows:

- A multi-agent scenario is designed where UAV swarms inspect telecommunication towers, considering practical constraints such as collision avoidance, task deadlines, and kinematic constraints.

- Mobile intelligent jammers capable of learning interference strategies are introduced. These jammers emit interference signals to reduce the signal-to-interference-plus-noise ratio for UAV swarms. UAVs lack information about the jammers, while jammers can partially observe UAVs.

- This paper carefully designs the state space and reward functions, adopts the MADDPG algorithm framework. The performance of the trained models is quantitatively evaluated using multiple metrics, demonstrating that the trained path planning models exhibit good resilience and robustness.

Furthermore, for UAVs, training data is entirely derived from observable data in the environment, meaning UAVs do not require extensive prior knowledge about maps and jamming attacks. Given the consideration of various practical constraints, the model is more realistic and applicable.

II. SYSTEM MODEL

We consider a scenario where UAVs inspect telecommunication towers, and there are malicious jammers sending jamming signals during the data transmission process. The UAVs need to establish connections with pre-deployed inspection modules on telecommunication towers and exchange information to complete the inspection of the telecommunication towers. See Fig. 1.

The scene this article is interested in is a cube, expressed as $\mathbb{C} : \mathbb{X} \times \mathbb{Y} \times \mathbb{Z}$. The space constraint can be expressed as $\mathbb{X} \triangleq [x_{min}, x_{max}]$, $\mathbb{Y} \triangleq [y_{min}, y_{max}]$, $\mathbb{Z} \triangleq [z_{min}, z_{max}]$. There are multiple no-fly zones (obstacles) in the scene, expressed as $\mathbb{N} : \mathbb{X}^N \times \mathbb{Y}^N \times \mathbb{Z}^N$.

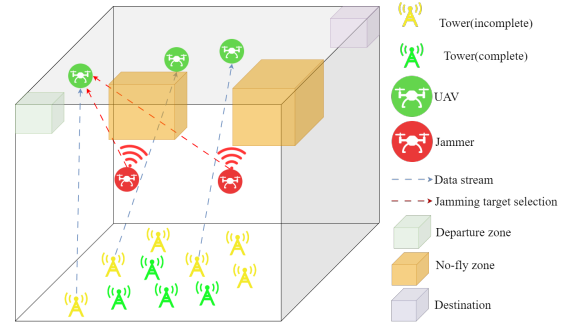


Fig. 1. UAVs need to complete communication between the inspection modules on telecommunication towers and themselves in the presence of jamming attacks. The jammer selects a UAV to send a jamming signal near it, but the jamming signal will affect all UAVs.

A. Communication Agents

- 1) *UAVs*. There are multiple cooperative UAVs in the scene. They need to collect log data from the inspection modules of telecommunication towers on the ground. The UAV is modeled as a sphere with radius r . The 3D position of the UAV $p^u = [p_x, p_y, H_u]$, where H_u is the flight height of the UAV. The information of UAV is represented as $s^u = [p^u, p^D, v^u, r, v_{max}^u] \in \mathbb{R}^{11}$. Where p^D represents the UAV mission destination, $v^u \triangleq [v_x, v_y]$ represents the current speed of the UAV, and v_{max}^u is the maximum flying speed of the UAV. The UAVs do not communicate with each other, so the strategies, motion status and other information of other UAVs are unknown to the current UAV. The assumption is that UAVs are equipped with low-cost sensors that can detect when other UAVs are approaching.
- 2) *Jammers*. The information of the jammer is similar to that of the UAV, but the jammer sends a jamming signal to affect the communication between the UAV and the inspection module. The interference signal power is P^j .
- 3) *Inspection Modules*. Since the actual communication involves the inspection modules of the telecommunication towers, which are located in fixed positions, these inspection modules will henceforth be referred to as nodes. There are N nodes in the scene located at a fixed position: $p^n = [p_{x_n}, p_{y_n}]$ on the ground. each node has b_n bits of log data to upload. The Node has two modes: 1) active mode when the data transmission is not completed, waiting for the UAV to establish a connection with itself or transmitting data at power P_n in the established connection; and 2) silent mode when the data transfer is complete.

B. Communication Channel

Due to the high altitude of the UAV, it is assumed that the communication between any agents is line-of-sight. Path loss can be expressed as:

$$L(d) = (d^2 + H^2)^{\alpha/2}, \quad (1)$$

where α is the path loss exponent; d is the horizontal distance between two communication agents; H is the vertical distance between two communicating agents. Assume that the omni-directional antenna gains of the nodes and jammers are $G_n = 0\text{dB} = G_J$. The UAV is equipped with a receiver with a horizontal directional antenna, and a simple analytical approximation of the antenna gain provided by the UAV can be expressed as:

$$G_u(d) = \sin(\theta) = \frac{d}{\sqrt{d^2 + H^2}}, \quad (2)$$

where θ is the elevation angle between two communication agents. Subsequent analysis is applicable to any type of antenna pattern, even specific antenna gains are considered.

The signal power of a UAV received from n^{th} node can be expressed as:

$$P_n^u = P^n G_u(d_n) L^{-1}(d_n), \quad (3)$$

thus, the signal-to-interference-plus-noise ratio (SINR) at the UAV if the connection is established can be expressed as:

$$S_n^u \triangleq \frac{P^n G_u(d_n) L^{-1}(d_n)}{\mathcal{N}_s + I^{ju}}, \quad (4)$$

where $\mathcal{N}_s$ is the noise power, I^{ju} is the jamming signal power sent by the jammer, expressed as:

$$I^{ju} = P^j G_u(d_{ju}) L^{-1}(d_{ju}), \quad (5)$$

where, P^j is the transmit power of the jammer, and d_{ju} is the horizontal distance between the UAV and the jammer.

When a UAV establishes with n^{th} node, the maximum reachable information rate is:

$$R_n^{\text{max}} = \log_2(1 + S_n^u). \quad (6)$$

To ensure the quality of communication, a reliable link must be established between the UAV and the node. In order to achieve this, a SINR threshold $\mathcal{S}^u$ is set in advance. If and only if the SINR S_n^u calculated by (4) is greater than the threshold, the UAV can communicate with the node. The formula for calculating the effective information rate obtained is as follows:

$$R_n^u = \begin{cases} R_n^{\text{max}}, & \log S_n^u > \mathcal{S}^u, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

We use the standard Time Division Multiple Access (TDMA) model. The UAV can only communicate with one node at a time. Using $q_n^V \in (0, 1)$ to represent the connection between the n^{th} node and the UAV:

$$\sum_n^N q_n^u \leq 1. \quad (8)$$

Scheduling is based on a maximum SINR strategy, the UAV communicates with the node that provides the maximum S_n^u . This strategy can be mathematically expressed as:

$$q_n^u = \begin{cases} 1, & \log n = \operatorname{argmax}_{n' \in \{\mathcal{N}\}} S_{n'}^u, \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

where $\mathcal{N}$ represents the activated nodes.

III. PROBLEM FORMULATION

This section first describes the task scenario with an optimization problem, then treats it as a multi-agent extension of the Markov decision process [19], and then gives our solution.

A. Problem Formulation for Attack

The goal of jammers' is to get as close to UAVs as possible at all times and reduce their SINR at any moment. Specifically, the optimization problem for jammers(attack) can be formulated as:

$$\operatorname{argmin}_{\{p_t^j \forall t, j \in \mathcal{J}\}} \mathbb{E} \left[\sum_{u \in \mathcal{U}} \sum_{n=1}^N \sum_{t=0}^T q_n^u \Delta t R_{nt}^u \right] \quad (10a)$$

$$\text{s.t. } \|p_t^j - p_t^i\|^2 > r^i + r^j, \quad \forall i, j \in \mathcal{J}, i \neq j \quad (10b)$$

$$T \Delta t \leq \mathcal{T} \quad (10c)$$

$$v_t^j \leq v_{\text{max}}^j, \forall t, \forall j \in \mathcal{J} \quad (10d)$$

where p_t^j is the position of Jammer j at time t . In (10b), collision avoidance constraints are given. (10c) stands for the mission completion deadline, which is the same as UAVs'. (10d) means the kinematic constraints. To exacerbate their interference with UAVs, Jammers are exempt from obstacle avoidance constraints. Additionally, Jammers are equipped with inexpensive sensors that enable them to detect and approach nearby UAVs. Furthermore, Jammers have access to continuous reference signal received power (RSRP) and reference signal received quality (RSRQ) reports from UAVs, allowing them to determine the UAVs' signal-to-interference-plus-noise ratio information.

It is worth noting that Jammers, who have access to this information, are strong adversaries. At the same time, there are fewer constraints on Jammers, which means that they can more easily carry out jamming attacks against UAVs. Therefore, this article considers the worst-case scenario. If UAVs can successfully defend against such powerful intelligent Jammers, it will be much more effective against other jammers.

We have noticed that when Jammers and UAVs are flying at the same altitude, Jammers need to get as close as possible to UAVs without colliding with it. In addition, when using a sine antenna pattern, if Jammers and UAVs are at the same height, UAVs will not receive interference from Jammers (or the interference is really small). So this article sets up Jammers and UAVs to fly at different altitudes.

B. Problem Formulation for Defence

The goal of UAVs is to design resilient paths that meet various practical constraints and effectively complete the data collection mission in the presence of jamming attacks. Similar to the Jammers' form. the UAVs' optimization problem can be formulated as:

$$\operatorname{argmax}_{\{p_t^u \forall t, u \in \mathcal{U}\}} \mathbb{E} \left[\sum_{u \in \mathcal{U}} \sum_{n=1}^N \sum_{t=0}^T q_n^u \Delta t R_{nt}^u \right] \quad (11a)$$

$$\begin{aligned} s.t. \quad & \|p_t^u - p_t^v\|^2 > r^u + r^v, \\ & \forall u, v \in \mathcal{U}, u \neq v \end{aligned} \quad (11b)$$

$$T \Delta t \leq \mathcal{T} \quad (11c)$$

$$v_t^u \leq v_{max}^u, \forall t, \forall u \in \mathcal{U} \quad (11d)$$

$$\sum_n q_n^u \leq 1, \forall u \in \mathcal{U} \quad (11e)$$

$$p_T^u = p^D \quad (11f)$$

$$p_t^u \notin \mathbb{N}, \forall t \quad (11g)$$

Compared to the Jammers' form, the UAVs' optimization problem has added the following: 1) (11e), which means TDMA constraint; 2) (11f), which means the destination locations constrain; 3) (11g), which means the no-fly zones constrain.

As described in III-A, there are more stringent constraints on UAVs.

C. Markov Game for Multi-UAV

In multi-agent environments, the decision of each agent affects the joint action of all agents. We model the environment as a multi-agent extension of the Markov decision process [19]. A multi-agent Markov game can be formalized as a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{R}$ is the reward space, $\mathcal{P}$ is the transition probability and γ is the reward discount factor.

At each time step t , each agent (for example agent i) observes the environment and gets a state $s_i^t \in \mathcal{S}$, and according to a deterministic policy, selects an action $a_i^t \in \mathcal{A}$, which leads to a new state s_i^{t+1} with transition probability $\mathcal{P}(s_i^{t+1} | s_i^t, a_1^t, \dots, a_N^t)$ (N stands for the number of agents). The discounted return of agent i is defined as $\mathcal{R}_i^t = \sum_{n=0}^L \gamma^n r_i^{t+n}$, where L is the time horizon.

In this paper, the Agent, State, Action, and Reward of multi-agent deep reinforcement learning models are defined as follows:

- 1) *Agent*. each UAV (including UAVs and Jammers) in the environment can be regarded as an agent. Each UAV has its own critic and actor networks, where the critic network takes the state-action joint vector of all cooperative UAVs as input during training and gives a score to the action given by the current actor.
- 2) *State*. Unlike related work [15], [18] that only includes basic position and velocity information of UAVs, this article incorporates richer information into the state space. Apparently, this method of preprocessing the state with

a feature engineering-like idea makes the information entering the neural network more effective and helps to improve the convergence speed of the algorithm. Moreover, these processes only require simple calculations, so they will not take up too many computing resources. Specifically, the design of the state space is as follows:

- a) *Jammers*. $s^j \triangleq [p_{[0:2]}^j, v^j, p^{tar}, d^{tar}, rnv^{tar}]$, where $p_{[0:2]}^j$ represents the two-dimensional coordinates of the jammer projected onto the ground (since the jammer flies at a fixed altitude, the altitude H^j is redundant); v^j represents the velocity component of the jammer along the coordinate axis; p^{tar}, d^{tar} and rnv^{tar} represent the relative two-dimensional coordinates, distance, and SINR of the current target UAV.
- b) *UAVs*. $s^u \triangleq [p_{[0:2]}^u, v^u, p^{tar}, flag_{cnt}, p^{cu}, p^{cj}]$, where $p_{[0:2]}^u, v^u$ are similar to Jammers'; p^{tar} represents the relative two-dimensional coordinates of the current target (a node, if it is currently connected, or the endpoint, otherwise); p^{cu}, p^{cj} represent the coordinates of the cooperative UAVs and Jammers that can be detected by the sensor and $flag_{cnt}$ represents the current connection status (1 for established connection, 0 otherwise).

- 3) *Action*. The action of each agent is the output of their respective actor network, which has a simple form: $[a_x, a_y]$, representing the acceleration components along the two coordinate axes.
- 4) *Reward*. When agent i takes action a to transit from state s_t to s_{t+1} , a reward r_i will be acquired. We take the average of all the reward values as the team reward. Next, the specific forms of the reward functions of Jammers and UAVs are given. It is assumed by default that all occurrences of $\alpha_*^{j/u}$ in the following text are positive constants.

- a) *Jammers*. According to the objective function and constraints of Jammers in III-A, the reward function is designed as follows:

$$\mathcal{R}_t^j = \mathcal{R}_{ct}^j + \mathcal{R}_{st}^j + \mathcal{R}_{dt}^j. \quad (12)$$

The first term $\mathcal{R}_{ct}^j$ represents the collision penalty. The second term $\mathcal{R}_{st}^j$ is the SINR piecewise reward term. The last term $\mathcal{R}_{dt}^j$ is the reward term for the distance to the current target.

- b) *UAVs*. According to the objective function and constraints of Jammers in III-B, the reward function is designed as follows:

$$\mathcal{R}_t^u = \mathcal{R}_{ct}^u + \mathcal{R}_{st}^u + \mathcal{R}_{tt}^u. \quad (13)$$

The first term $\mathcal{R}_{ct}^u$ represents the penalty for flight constraints. The second term $\mathcal{R}_{st}^u$ is a reward term that encourages the UAV to improve the SINR. The last term $\mathcal{R}_{tt}^u$ is a reward item that encourages the UAV to go to the destination after all the nodes

have finished transmitting, and it is designed as a potential function.

In our setup, the UAVs and the Jammers are considered as two separate cooperative agent teams, and they are trained separately.

IV. SIMULATION RESULTS

In this section, the specific implementation details of the scenarios demonstrated are presented, and the performance of the anti-jamming resilient path planning models obtained by the three algorithms is demonstrated.

Inspired by [15], four metrics are designed to evaluate the performance of the trained model. 1) **transmission Success Rate (SR)**, which represents the proportion of nodes that successfully complete data transmission; 2) **Data collection Rate (DR)**, which represents the proportion of log data collected when the task is aborted; 3) **Collision Rate (CR)**, which represents the proportion of time steps in which a collision occurs; 4) **no-fly zone Intrusion Rate (IR)**, which represents the proportion of time steps in which a no-fly zone intrusion occurs.

A. Environment Settings

We consider a cubic space of $10m \times 10m \times 10m$. UAVs fly at a height of $H^u = 10m$, jammers fly at a height of $H^j = 5m$, and nodes are located on the ground. The transmit power of the node is $p = 0.1w$, and the transmit power of the Jammers is $p^j = 0.001/3w$. There are two no-fly zones in the scene: $NF_1 : [x_{min} = 5.5m, x_{max} = 7.5m, y_{min} = 5.5m, y_{max} = 7.5m]$, $NF_2 : [x_{min} = 2.5m, x_{max} = 3.5m, y_{min} = 2.5m, y_{max} = 3.5m]$. We consider three UAVs with starting points $p_1^U = [0, 0, 10]$, $p_2^U = [0, 5, 10]$ and $p_3^U = [5, 0, 10]$, and two Jammers with starting points $p_1^J = [7, 3, 5]$ and $p_2^J = [4, 9, 5]$. The destination of all UAVs is $p^D = [10, 10, 10]$. The maximum velocity components of UAVs and jammers along the x-axis and y-axis are $5m/s$, and the maximum acceleration is $10m/s^2$. The radius of the UAV is $r = 0.2m$. The task time constraint is $20s$, and each control time step is $0.05s$. The path loss is $\alpha = 1.5$, and the SINR threshold during training is $\mathcal{S}^u = 3$.

B. Training Settings

Both the actor networks of UAVs and jammers are designed as three-layer DNN of (256,128,128), and the critic network is designed as a two-layer DNN of (256,256). Tanh and leaky relu activation functions are used according to different requirements in the neural network. Both critic and actor networks use the Adam optimizer to update parameters. The learning rate for the critic network is $lr_c = 0.005$, and the learning rate for the actor network is $lr_a = 0.0005$.

C. No Jamming Attack Scenario

Firstly, a no jamming attack scenario is designed, scenario, where UAVs only need to collect log data from the nodes as close as possible and arrive at the destination as early as possible. MADDPG can converge quickly in this scenario.

Fig. 2 shows the trajectories in this scenario, among them, the black, green, and blue paths are the flight paths of the three UAVs, and the red paths are the flight paths of the two jammers. The rectangles in the figure represent the no-fly zones, and the triangles are the nodes. As can be seen from Table I, UAVs are capable of successfully completing their tasks with a very low error rate.

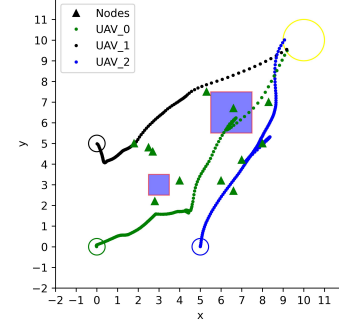


Fig. 2. Model trajectories trained and tested in the absence of jamming attacks.

D. Fixed Jamming Attack Scenario

Next, fixed jammers are added to the scenario. They are located at fixed locations, and reduce the overall communication between UAVs and nodes. Under this setting, the model obtained in the previous subsection will be greatly affected. It can be seen that the jamming attack can significantly reduce the model performance. At the same time, model training was conducted under this setting, and the obtained model can well resist this non-intelligent attack. Fig. 4 depicts the trajectories plot of the UAVs that have been trained and tested under this setting.

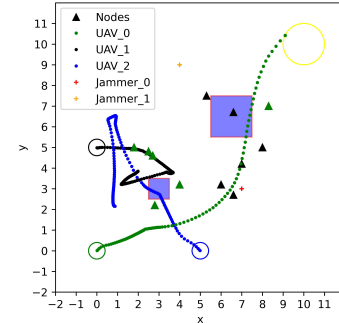


Fig. 3. Model trajectories trained in an environment without jamming attacks and tested in an environment with fixed jamming nodes.

E. Intelligent Jamming Attacks and Defenses with MADDPG Training

Under this setting, UAVs and Jammers are considered as two separate cooperative teams, each trained with the MADDPG algorithm. In order to verify the resilience of the model, the positions of the nodes were randomized during the test. It is

TABLE I
PERFORMANCE OF DIFFERENT MODELS ON DIFFERENT TEST ENVIRONMENTS

Algorithm		Train environment	Test environment	SR(%)	DR(%)	CR(%)	IR(%)
UAVs	Jammers						
MADDPG	None	No jammer	No jammer	100	100	2.00	3.08
MADDPG	None	No jammer	Fixed jammers	91.67	91.92	0.00	0.40
MADDPG	None	Fixed jammers	Fixed jammers	100	100	5.00	0.17
MADDPG	MADDPG	Intelligent jammers	Intelligent jammer	99.67	99.68	0.07	1.21

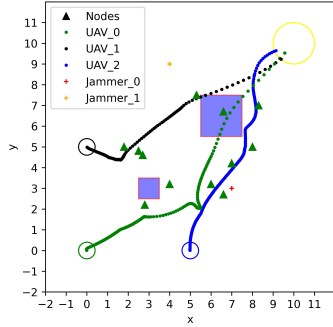


Fig. 4. Model trajectories trained and tested in the presence of fixed jamming nodes.

intriguing that in the context of team-versus-team scenarios, when the number of training episodes is sufficiently large, the UAVs exhibit intelligent team behaviors. One UAV is designated to attract Jammers, thus ensuring that its two teammates can accomplish their tasks more efficiently.

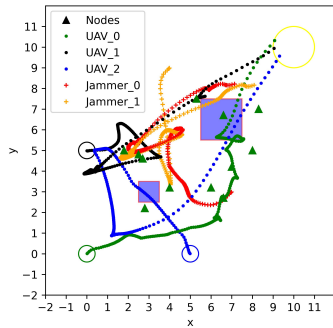


Fig. 5. Model trajectories trained by MADDPG to defense intelligent jamming attacks.

REFERENCES

- [1] Y. Zeng, Q. Wu, and R. Zhang, "Accessing from the sky: A tutorial on uav communications for 5g and beyond," in *Proceedings of the IEEE*, vol. 107, no. 12, pp. 2327–2375, 2019.
- [2] H. Menouar, I. Guvenc, K. Akkaya, A. S. Uluogac, A. Kadri, and A. Tuncer, "UAV-enabled intelligent transportation systems for the smart city: Applications and challenges," *IEEE Communications Magazine*, vol. 55, no. 3, pp. 22–28, 2017.
- [3] K. Harikumar, J. Senthilnath, and S. Sundaram, "Multi-UAV oxyrrhis marina-inspired search and dynamic formation control for forest fire-fighting," *IEEE Transactions on Automation Science and Engineering*, vol. 16, no. 2, pp. 863–873, 2019.
- [4] H. Feng, J. Wang, Z. Fang, J. Chen, and D.-T. Do, "Evaluating AoI-centric HARQ protocols for UAV networks," *IEEE Transactions on Communications*, vol. 72, no. 1, pp. 288–301, 2024.
- [5] M. F. M. Isa, N. Z. A. Rahim, and M. S. Fathi, "It's a bird... It's a plane... It's a drone...: Telecommunication Tower Inspection Using Drone," in *Proceedings of International Conference on Research and Innovation in Information Systems*, 2019, pp. 1–5.
- [6] D. Wang, B. Bai, W. Zhao, and Z. Han, "A survey of optimization approaches for wireless physical layer security," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 2, pp. 1878–1911, 2018.
- [7] D. Darsena, G. Gelli, I. Iudice, and F. Verde, "Detection and blind channel estimation for UAV-aided wireless sensor networks in smart cities under mobile jamming attack," *IEEE Internet of Things Journal*, vol. 9, no. 14, pp. 11 932–11 950, 2021.
- [8] M. F. Ahmed, J. Mohanta, A. Sanyal, and P. S. Yadav, "Path planning of unmanned aerial systems for visual inspection of power transmission lines and towers," *IETE Journal of Research*, pp. 1–21, 2023.
- [9] B. Duan, D. Yin, Y. Cong, H. Zhou, X. Xiang, and L. Shen, "Anti-jamming path planning for unmanned aerial vehicles with imperfect jammer information," in *Proceedings of IEEE International Conference on Robotics and Biomimetics*. IEEE, 2018, pp. 729–735.
- [10] B. Duo, Q. Wu, X. Yuan, and R. Zhang, "Anti-jamming 3D trajectory design for UAV-enabled wireless sensor networks under probabilistic LoS channel," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 12, pp. 16 288–16 293, 2020.
- [11] Y. Wu, W. Fan, W. Yang, X. Sun, and X. Guan, "Robust trajectory and communication design for multi-UAV enabled wireless networks in the presence of jammers," *IEEE Access*, vol. 8, pp. 2893–2905, 2019.
- [12] H. Ye, G. Y. Li, and B.-H. F. Juang, "Deep reinforcement learning based resource allocation for V2V communications," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 4, pp. 3163–3173, 2019.
- [13] Z. Cao, H. Guo, W. Song, K. Gao, Z. Chen, L. Zhang, and X. Zhang, "Using reinforcement learning to minimize the probability of delay occurrence in transportation," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 3, pp. 2424–2436, 2020.
- [14] C. Han, A. Liu, K. An, H. Wang, G. Zheng, S. Chatzinotas, L. Huo, and X. Tong, "Satellite-assisted UAV trajectory control in hostile jamming environments," *IEEE Transactions on Vehicular Technology*, vol. 71, no. 4, pp. 3760–3775, 2021.
- [15] X. Wang and M. C. Gursoy, "Resilient path planning for UAVs in data collection under adversarial attacks," *IEEE Transactions on Information Forensics and Security*, 2023.
- [16] F. Tang, Z. M. Fadlullah, N. Kato, F. Ono, and R. Miura, "AC-POCA: Anticoordination game based partially overlapping channels assignment in combined UAV and D2D-based networks," *IEEE Transactions on Vehicular Technology*, vol. 67, no. 2, pp. 1672–1683, 2017.
- [17] R. Lowe, Y. I. Wu, A. Tamar, J. Harb, O. Pieter Abbeel, and I. Mordatch, "Multi-agent actor-critic for mixed cooperative-competitive environments," *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [18] Y. Zhang, Z. Mou, F. Gao, J. Jiang, R. Ding, and Z. Han, "UAV-enabled secure communications by multi-agent deep reinforcement learning," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 10, pp. 11 599–11 611, 2020.
- [19] M. L. Littman, "Markov games as a framework for multi-agent reinforcement learning," in *Machine Learning Proceedings*. Elsevier, 1994, pp. 157–163.